Supporting Information for the IEC Res Manuscript:

Phase Equilibria Of CO_2 + n-Alkane Binary Systems In Wide Ranges of Conditions: Development of Predictive Correlations Based on Cubic Mixing Rules

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Description:

This document contains Appendix A, which presents a literature review on the experimental fluid phase equilibria of binary carbon dioxide + n-alkane systems; and Appendix B, which presents a number of tables refered to in the main text of the article.

APPENDIX A: Experimental Data on the fluid phase equilibria of Carbon Dioxide + n-Alkane systems

We performed a literature review that focused on the experimental fluid phase equilibria of binary carbon dioxide + n-alkane systems, where the n-alkane of the mixture is a linear chain molecule having 1 to 36 carbon atoms. Equilibria with presence of solid phases have not been taken into account. The literature on the experimental phase behavior of the CO_2 + n-alkane homologue series is vast. Although we have identified an important number of relevant contributions, it is unlikely for our final list in the present work to be complete.

The information on CO_2 + n-alkane fluid phase equilibria is arranged in tables A1 to A6. A given table corresponds to a specific type of phase equilibrium information. The least familiar type is probably the “critical end point” (CEP). A CEP is a point where a line of binary critical points

(critical line) terminates. At a CEP a critical phase is at equilibrium with a non critical phase (see reference[1] for more details) . At a so-called "K" CEP the critical phase corresponds to a vapor-liquid critical point.

The tables have as main entry the carbon number (CN) of the n-alkane of the studied CO_2 + n-alkane mixture. All tables report the method employed to obtain the experimental data, according to Deiters and Schneider[2], who classify the methods, considering their underlying principles, into synthetic and analytic. In the synthetic methods, a mixture of known composition is prepared, and then its phase behavior is observed as a function of temperature and/or pressure. In the analytic methods, the phase separation is promoted through changes in temperature and/or pressure (as in the synthetic methods); then samples of the phases are withdrawn an analyzed through appropriate methods. Furthermore, we consider an additional criterion for the classification of the experimental methods; we assort them as static or dynamic methods. In the former case, none of the phases of the system flows through the equilibrium cell during the measurements. In the latter case, one or more phases are allowed to flow during the measurement to promote the equilibrium condition inside the equilibrium cell. For more details, see reference[3]. Other possible classifications have been proposed[3,4,5,6,7].

Tables A1 to A6 also report the purity of the compounds in the mixture together with the corresponding units. Notice that the last column in each table identifies the reference from which the data were taken.

Table A1 presents information on literature contributions reporting experimental critical end points (CEPs) for CO_2 + n-alkane binary mixtures, and it also reports the temperature and pressure coordinates of such CEPs. According to Fall and Luks[8], binary systems formed by carbon dioxide and n-alkanes up to C6 exhibit a type I behavior according to the classification proposed by Scott

and van Konynenburg[9] for the fluid phase behavior of binary mixtures([1]). This means that a liquid=vapor (L=V) critical locus extends from the critical point of the pure n-alkane to the critical point of pure carbon dioxide, and there are no CEPs. The binary systems containing CO_2 and n-alkanes from C7 to C12 display a type II phase behavior[9]. In type II behavior, a critical locus extends between the critical points of the pure compounds, as in type I behavior, and another critical locus (liquid=liquid, L=L) appears, which finishes in an upper critical end point (UCEP). A liquid-liquid-vapor (LLV) locus goes from that UCEP to lower temperatures. The system formed by CO_2 and n-Tridecane constitutes a "transition" from type II to type III[9] phase behavior in this homologue series, and it presents a type IV behavior. In type IV phase behavior, the UCEP which signals the end of the L=L critical locus is also present, as in type II behavior. However, the critical line that stems from the critical point of the heavy compound, i.e., the n-alkane, ends at a lower critical end point (LCEP); and another critical locus extends from the critical point of pure CO_2 to a second UCEP (these last two types of critical lines were denoted E and D respectively, by Cismondi and Michelsen[10]). A LLV locus connects the last two CEPs. Finally, as already pointed out, for example, by Fall and Luks[8], binary mixtures made of CO_2 and n-alkanes, with CN greater than or equal to 14, show a type III phase behavior. In this type of behavior a critical locus connects the critical point of pure CO_2 with an UCEP, while there exists another critical locus that originates at the critical point of the pure n-alkane and extends indefinitely to high pressures. This latter critical locus (denoted C by Cismondi and Michelsen[10]) may show local extrema in the pressure vs. temperature plane. As the difference between the critical temperatures of pure carbon dioxide and the pure n-alkane increases, the asymmetry of the mixture rises up, and we observe an evolution in the fluid phase behavior for this homologue series. This kind of evolution has also been observed for other homologue series[1].

([1]) Actually, it could be argued that the fluid phase behavior of these systems corresponds to type II, but with the LLV región and the Upper CEP (UCEP) subject to the interference of the more stable solid-fluid equilibria.

Table A2 summarizes information on critical point experimental data sets reported in the literature for CO_2 + n-alkane binary mixtures. Table A2 gives, for each record, the temperature and pressure ranges for the measured critical loci, together with the temperature and pressure uncertainty estimates. It also indicates whether the experimental critical composition, density, and composition uncertainty, are available. "N/A" or "NO" in Tables A1 to A6 means a lack of information for the corresponding field in the quoted reference (see last column in Tables A1 to A6). "YES" means availability of some information for the corresponding field.

Table A3 reports the literature sources of liquid-liquid-vapor (LLV) experimental data for CO_2 + n-alkane binary mixtures. For each record in Table A3, we provide information similar in nature to that given in Table A2. Table A3 includes an additional field ("Phases informed") that identifies the phases for which the authors have reported the experimental composition and/or density. "L1" is the liquid phase richer in the n-alkane, "L2" is the liquid phase richer in CO_2, and "V" is the vapor phase, being the phases "L1", "L2" and "V" in equilibrium.

From the data depicted in Table A3, we observe that there is a lack of LLV experimental information for some binary systems in this homologue series, as the ones corresponding to C7, C9, C11, C17 and C18. For some of the systems here classified as type III (CN>13), corresponding to n-Docosane and heavier n-alkanes, the LLV locus is interrupted by the appearance of a quadruple point (Q point) [11, 12] where three fluid phases are at equilibrium with a solid phase. Since the pure n-alkane melting point increases with the n-alkane molecular weight, the Q point temperature of the binary CO_2 + n-alkane system also increases with it.

Table A4 presents information on literature contributions reporting fluid phase equilibrium data measured at constant temperature for CO_2 + n-alkane mixtures. In this table we inform the

temperature of the isothermal set, the pressure range of the measured data, the availability or not of the compositions and composition uncertainties of the phases at equilibrium (n-alkane richer and CO_2 richer phases), and the uncertainties in temperature and pressure, if available. Besides, we include an additional field stating whether the experimental density of the mixture has been reported in the corresponding reference.

The length of Table A4 clearly shows that isothermal two-phase equilibrium is the type of experimental data most frequently gathered and reported for the CO_2 + n-alkane series. Often, specially for the heavier n-alkanes, some important degree of scatter is observed when graphically comparing data sets at the same temperature from different literature sources.

Moreover, even serious disagreements can be found, as illustrated in Fig. 1 for CO_2 + n-Hexadecane at 313.2 K (see the main body of this work).

Table A5 gives information on experimental data sets at constant system pressure (Txy sets) available in the literature for two-phase fluid-fluid equilibria of CO_2 + n-Alkane binary mixtures.

Table A6 presents information on experimental fluid-fluid equilibria isoplethic data sets available in the literature for CO_2 + n-alkane binary mixtures. For a given data set, Table A6 reports the isopleth composition, and the temperature and pressure ranges of the experimental data. Table A6 also informs the composition, temperature and pressure uncertainties. It also indicates whether the density of the mixture has been reported in the corresponding reference.

We stress that the most often reported experimental phase equilibrium data sets, in the present collection, are of the isothermal type (see Tables A4 to A6). We observe that, even though the

homologue series CO_2 + n-alkane has been widely studied, there is still some lack of important experimental information (see, e.g., Table A3).

Table A1. Experimental data on fluid-fluid critical end points for CO_2 + n-Alkane binary mixtures

CN	Method	Purity		Units for Purity		UCEP				LCEP				UCEP				Reference
		CO_2	Alk.	CO_2	Alk.	T (K)	Absolute T uncertainty (K)	P (bar)	Absolute P uncertainty (bar)	T (K)	Absolute T uncertainty (K)	P (Bar)	Absolute P Uncertainty (bar)	T (K)	Absolute T Uncertainty (K)	P (Bar)	Absolute P Uncertainty (bar)	
7	Analytic static	99.9	99.9	Mole %	Mole %	222.6	0.2	6.5	0.03									Im and Kurata, 1971[13]
8	Analytic static	99.99	99.0	Mole %	Mole %	231.5	0.03	9.2	0.07									Hottovy et al., 1982[14]
10	Static synthetic	> 99.99	>99.0	%	%	248.7	0.15	16.4	0.07									Kulkarni et al., 1974[15]
12	Analytic static	99.99	99	Mole %	Mole %	267.3	0.02	28.8	0.07									Hottovy et al., 1981[16]
13	Analytic static	99.99	99	Mole %	Mole %	279.0	0.02	39.3	0.07									Hottovy et al., 1981[16]
13	Analytic static	99.99	99	Mole %	Mole %					310.8	0.02	81.1	0.07	314.0		87.2		Fall and Luks, 1985[8]
14	Analytic static	99.99	99	Mole %	Mole %									311.2	0.02	82.6	0.07	Hottovy et al., 1981[16]
15	Analytic static	99.99	99	Mole %	Mole %									309.4	0.02	80.2	0.07	Hottovy et al., 1981[16]
19	Analytic static	99.99	99	Mole %	Mole %									305.5		75.8		Fall et al. 1985[17]
20	Analytic static	99.99	99	Mole %	Mole %									305.3	0.02	75.5	0.07	Hottovy et al., 1981[16]
20	Analytic static	99.99	99	Mole %	Mole %									305.2		75.4		Fall et al. 1985[17]
21	Analytic static	99.99	99	Mole %	Mole %									305.0		75		Fall et al. 1985[17]

Table A2. Data sets of experimental critical points available for CO_2 + n-Alkane binary mixtures

CN	Method	Purity		Units for Purity		T range (K)	Absolute T uncertainty (K)	P range (bar)	Absolute P Uncertainty (bar)	Composition informed	Composition uncertainty	Density informed	Reference
		CO_2	Alk	CO_2	Alk								
1	DA	99.99	99.99	%	%	193.1-219.3	0.01	47.4-64.9	0.7	YES	YES	NO	Mraw et al, 1978[18]
2	DS	0.99995	0.9995	Mass fr.	Mass fr.	291-305.4	0.5	57.11-48.17	0.5	YES	YES	NO	Horstmann et al, 2000[19]
3	SA	99.5	99.9	%	%	304.4-366.2	0.3	54.5-70.5	0.3	YES	YES	NO	Poettmann and Katz, 1945[20]
3	SA	99.95	98	% vol.	% vol.	304.3.-364.8	N/A	47.4-70.7	N/A	YES	NO	NO	Smejkal et al, 2002[21]
3	SA	99.95	98	% vol.	% vol.	327-369.7	N/A	42.5-65.2	N/A	YES	NO	NO	Smejkal et al, 2002[21]
4	SA	99.5	99.0	%	%	321.8-415.8	0.3	48.9-81.4	0.3	YES	YES	NO	Poettmann and Katz, 1945[20]
4	SA	0.999	0.999	Mole fr.	Mole fr.	325.9-412.3	0.06	51.1-81.7	0.01	YES	YES	YES	Olds et al., 1949[22]
4	DA	99.99	99.98	%	%	319.3-377.6	N/A	75.7-81.2	N/A	YES	NO	YES	Hsu et al, 1985[23]
4	SA	99.9	99.9	%	%	368.15-418.15	0.1	46.4-77.98	0.1	NO	YES	NO	Leu and Robinson, 1987[24]
5	SA	99.5	99.0	%	%	314.8-404.8	0.3	79.2-99.4	0.3	YES	YES	NO	Poettmann and Katz, 1945[20]
5	SS	99.995	>99.8	%	%	312.35- 326.15	0.05	77.5- 85.2	0.25	Yes		Yes	Chen et al, 2003[25]
5	SA	99.9	99.4	%	%	408.1-463.1	0.1	45.2-81.8	0.1	NO	YES	NO	Leu and Robinson, 1987[26]
6	SS	99.9	99.9	%	%	313.6-351.8	0.1	80.4-102.0	0.6	YES	YES	NO	Choi and Yeo, 1998[27]
6	SS	> 99.5	> 99.5	Mole %	Mole %	304.5-507.4	0.3	30.2-117.2	0.3	YES	YES	NO	Liu et al., 2003[28]
7	SA	> 99.9	> 99.9	Mole %	Mole %	310.6-477.2	0.18	77.9-137.1	0.21	NO	YES	NO	Kalra et al., 1978 [29]
7	SS	99.9	99.9	%	%	324.0-364.3	0.1	93.0-118.8	0.6	YES	YES	NO	Choi and Yeo, 1998[27]
7	DS	N/A	N/A	N/A	N/A	304.2-620.1	N/A	27.0-133.0	0.5	NO	NO-	NO	Chester and Haynes, 1997[30]
8	SS	99.9	99.9	%	%	329.4-359.5	0.1	101.2-130.3	0.6	YES	YES	NO	Choi and Yeo, 1998[27]
9	SS	99.9	99.9	%	%	342.3-386.8	0.1	119.2-148.0	0.6	YES	YES	NO	Choi and Yeo, 1998[27]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	304.1-619.3	0.09	21.0-176.8	0.01	YES	YES	YES	Reamer and Sage, 1963[31]
10	DS	Reagent grade	SFC/SFE grade	N/A	N/A	304.2-620.1	N/A	21.1-185	0.5	NO	NO	NO	Chester and Haynes, 1997[30]
10	DA	99.99	99.0	%	%	344.3-377.6	N/A	127.2-164.9	N/A	YES	NO	YES	Nagarajan and Robinson , 1986[32]
13	SS	99.99	99.0	%	%	300.8-377.4	0.25	67.5-198.0	0.12	YES	NO	NO	Enick et al., 1985[33]
13	SS	99.99	99.0	%	%	310.8-313.0	0.25	82.77-86.32	0.12	YES	NO	NO	Enick et al., 1985[33]
14	SS	N/A	N/A	N/A	N/A	283.1-393.5	N/A	78.2- 997.8	N/A	NO	NO	NO	Scheidgen, 1997[34]
14	SA	N/A	N/A	N/A	N/A	283.0-353.0	N/A	80.0-215.0	N/A	NO	NO	NO	Scheidgen and Schneider, 2000[35]
15	SA	N/A	N/A	N/A	N/A	290.0-353.0	N/A	120.0-210.0	N/A	NO	NO	NO	Scheidgen and Schneider, 2000[35]
15	SA	N/A	N/A	N/A	N/A	283-393	0.1	100-1000	1	NO	NO	NO	Pöhler et al. 1996[36]
15	SS	N/A	N/A	N/A	N/A	290.0-387.66	N/A	120.9-995	N/A	NO	NO	NO	Scheidgen, 1994[37]
16	SS	N/A	N/A	N/A	N/A	297.6-389.5	N/A	165.2-991.7	N/A	NO	NO	NO	Scheidgen, 1994[37]
16	DA	N/A	N/A	N/A	N/A	308.1-343.1	N/A	122.9-258.7	N/A	YES	NO	NO	Charoensombut-Amon et al, 1986 [38]
16	SA	N/A	N/A	N/A	N/A	293-393	0.1	100-1000	1	NO	NO	NO	Pöhler et al. 1996[36]

Table A2. Data sets of experimental critical points available for CO_2 + n-Alkane binary mixtures													
CN	Method	Purity		Units for Purity		T range (K)	Absolute T uncertainty (K)	P range (bar)	Absolute P Uncertainty (bar)	Composition informed	Composition uncertainty	Density informed	Reference
		CO_2	Alk	CO_2	Alk								
16	SA	N/A	N/A	N/A	N/A	295.0-393.0	N/A	150.0-100.0	N/A	NO	NO	NO	Scheidgen and Schneider, 2000[35]
16	SA	N/A	>99.5	N/A	%	393.2	0.2	256	2.0	YES	YES	NO	Spee and Schneider, 1991[39]
17	SS	N/A	N/A	N/A	N/A	304.7-393.1	N/A	203.3-991.1	N/A	NO	NO	NO	Scheidgen, 1997[34]
17	SS	N/A	N/A	N/A	N/A	323.2-393.2	N/A	213-273	N/A	NO	NO	NO	Pöhler, 1994[40]
18	SS	N/A	N/A	N/A	N/A	333.2-393.2	N/A	240-287	N/A	NO	NO	NO	Pöhler, 1994[40]
18	SA	N/A	N/A	N/A	N/A	323-393	0.1	235-290	1	YES	NO	NO	Pöhler et al. 1996[36]
19	SS	N/A	N/A	N/A	N/A	317.2-393.5	N/A	269.2-995.4	N/A	NO	NO	NO	Scheidgen, 1997[34]
19	SS	N/A	N/A	N/A	N/A	318.2-348.2	N/A	275-455	N/A	NO	NO	NO	Pöhler, 1994[40]
19	SA	N/A	>99.0	N/A	%	353.2-393.2	0.1	275.0-303.0	1.0	YES	NO	NO	Kordikowski and Schneider 1993[41]
20	SA	N/A	>99.0	N/A	%	353.2-393.2	0.1	302.0-320.0	1.0	YES	NO	NO	Kordikowski and Schneider 1993[41]
22	SS	N/A	N/A	N/A	N/A	334.2-392.3	N/A	350.3-991.0	N/A	NO	NO	NO	Scheidgen, 1997[34]
22	SS	N/A	N/A	N/A	N/A	393.2	N/A	357	N/A	NO	NO	NO	Pöhler, 1994[40]
22	SA	N/A	99	N/A	N/A	343.2-393.2	0.1	370-430	1	YES	NO	NO	Pöhler and Schneider 1995[42]
24	SA	N/A	>99.0	N/A	%	393.2	0.1	398.0	1.0	YES	NO	NO	Kordikowski and Schneider 1993[41]

SA: Static analytic. SS: Static synthetic. DA: Dynamic Analytic. DS: Dynamic synthetic. Mole fr.: Mole fraction. Mass fr.: Mass fraction

Table A3. Data Sets of Experimental Liquid-Liquid-Vapor equilibrium (LLV) points available for CO_2 + n-Alkane binary mixtures.													
CN	Method	Purity		Units for Purity		T range (K)	Absolute T uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Phases informed	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk								
8	SA	99.99	99.0	Mole %	Mole %	216.0-231.1	0.03	5.1-9.1	0.071	L1	YES	YES	Hottovy et al., 1982[14]
8	SA	99.99	99.0	Mole %	Mole %	217.2-231.0	0.03	5.3-9	0.071	L2	YES	YES	Hottovy et al., 1982[14]
10	SS	> 99.99	>99.0	%	%	235.6-248.7	0.15	10.6-16.4	0.07	L1-L2	NO	YES	Kulkarni et al, 1974[15]
12	SA	99.99	99	Mole %	Mole %	254.3-267.3	0.02	20.1-28.8	0.07	L1-L2	NO	YES	Hottovy et al. 1981[16]
13	SS	99.99	99.0	%	%	310.8-313.0	0.25	82-85.5	0.12	L1-L2	NO	YES	Enick et al., 1985[33]
13	SA	99.99	99	Mole %	Mole %	255.2-279.0	0.02	20.8-39.3	0.07	L1-L2	NO	YES	Hottovy et al. 1981[16]
13	SA	99.99	99	Mole %	Mole %	310.7-314.0	0.02	81.1-87.2	0.07	L1-L2	YES	YES	Fall and Luks 1985[8]
14	SA	99.99	99	Mole %	Mole %	269.1-311.1	0.02	30.8-82.6	0.07	L1-L2	NO	YES	Hottovy et al. 1981[16]
14	SA	99.99	99	%	%	279.3-303.1	0.03	39.8-69.2	0.1	L1-L2-V	YES	YES	Van der Steen et al. 1989[43]
14	ISA	99.995	N/A	Volume %	N/A	290-300	0.1	51.9-54.9	0.2	L1-L1	NO	YES	Laugier et al., 1990[44]
15	SA	99.99	99	Mole %	Mole %	270.4-309.4	0.02	31.9-80.2	0.07	L1-L2	NO	YES	Hottovy et al. 1981[16]
15	SA	99.99	99	%	%	273.4-308.4	0.03	34.5-78.4	0.1	L1-L2-V	YES	YES	Van der Steen et al. 1989[43]
15	SA	0.997	0.99	Mass fraction	Mass fraction	298.15	0.1	62.8	1.25	L1-L2-V	YES	NO	Secuianu et al., 2007[45]
15	SA	0.997	0.99	Mass fraction	Mass fraction	292.8-309.1	0.1	55.3-80.2	0.1	N/A	YES	NO	Secuianu et al., 2010[46]
16	SA	99.99	98	%	%	283.2-306.1	0.03	44.4-75.5	0.1	L1-L2-V	YES	YES	Van der Steen et al. 1989[43]
19	SA	99.99	99	%	%	292.9-305.5	0.02	56.7-75.7	0.07	L1-L2	YES	YES	Fall et al. 1985[17]
20	SA	99.99	99	Mole %	Mole %	300.4-305.2	0.05	67.5-75.5	0.07	L1-L2	YES	YES	Huie et al. 1973[47]
20	SA	99.99	99	%	%	300.3-305.2	0.02	67.3-75.4	0.07	L1-L2	YES	YES	Fall et al. 1985[17]
21	SA	99.99	99	%	%	301.5-305	0.02	69.2-75.1	0.07	L1-L2	YES	YES	Fall et al. 1985[17]

SA: Static analytic. ISA: Indirect static analytic. SS: Static synthetic.

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures														
CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
1	SA	99.99	99.99	%	%	153.1	0.01	11.5-11.6	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.99	99.99	%	%	173.1	0.01	24.7-26.0	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.99	99.99	%	%	183.1	0.01	33.6-36.4	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.99	99.99	%	%	193.1	0.01	42.9-47.4	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.99	99.99	%	%	203.1	0.01	47.6-52.7	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.99	99.99	%	%	210.1	0.01	41.6-57.5	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.99	99.99	%	%	219.3	0.01	5.7-63.8	0.07	YES	YES	YES	NO	Mraw et al, 1978[18]
1	SA	99.995	99.99	%	%	230	0.01	8.9-71.4	0.1	YES	YES	YES	NO	Webster and Kidnay, 2001[48]
1	SA	99.995	99.99	%	%	270	0.01	32-84.2	0.1	YES	YES	YES	NO	Webster and Kidnay, 2001[48]
1	DA	N/A	N/A	N/A	N/A	270	0.02	31.6-83.2	0.0015	YES	YES	YES	NO	Somait and Kidnay,1978[49]
1	SA	99.97	99.95	%	%	288.5	0.01	50.5-80.4	0.2	YES	YES	YES	NO	Xu et al, 1992[50]
1	SA	99.97	99.95	%	%	293.4	0.01	56.5-78.8	0.2	YES	YES	YES	NO	Xu et al, 1992[50]
1	DA	99.99	99.97	%	%	230	0.02	8.9-69.9	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
1	DA	99.99	99.97	%	%	250	0.02	17.8-80.5	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
1	DA	99.99	99.97	%	%	270	0.02	32.1-83.8	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
1	DA	99.991	99.97	%	%	219.3	0.02	5.81- 38.61	0.14	YES	YES	YES	No	Al-Sahhaf et al, 1983[52]
1	DA	99.991	99.97	%	%	240	0.02	12.87- 77.72	0.14	YES	YES	YES	No	Al-Sahhaf et al, 1983[52]
1	DA	99.991	99.97	%	%	270	0.02	32.03- 85.11	0.14	YES	YES	YES	No	Al-Sahhaf et al, 1983[52]
1	DA	N/A	99.97	N/A	%	230	0.01	8.9-71.75	0.05	YES	YES	YES	NO	Davalos et al., 1976[53]
1	DA	N/A	99.97	N/A	%	250	0.01	17.95-81.77	0.05	YES	YES	YES	NO	Davalos et al., 1976[53]
1	DA	N/A	99.97	N/A	%	270	0.01	31.94-85.93	0.05	YES	YES	YES	NO	Davalos et al., 1976[53]
2	DA	99.99	99	%	%	207	0.02	2.9-4.7	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
2	DA	99.99	99	%	%	210	0.02	3.3-5.5	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
2	DA	99.99	99	%	%	213	0.02	3.8-6.2	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
2	DA	99.99	99	%	%	230	0.02	7.0-11.5	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
2	DA	99.99	99	%	%	250	0.02	13.3-21.3	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
2	DA	99.99	99	%	%	270	0.02	22.1 -35.7	0.35	YES	YES	YES	NO	Wei et al, 1995[51]
3	SS	> 99.9	> 99.8	%	%	294.3	0.01	22.1-47.1	0.001	YES	NO	YES	NO	Kaminishi et al, 1987[54]
3	SA	99.995	99.5	%	%	230	0.01	0.97-7.8	0.1	YES	YES	YES	NO	Webster and Kidnay, 2001[48]
3	SA	99.995	99.5	%	%	270	0.01	4.3-92.9	0.1	YES	YES	YES	NO	Webster and Kidnay, 2001[48]
3	DA	99.99	99.95	%	%	240	0.01	1.5-12.7	0.1	YES	YES	YES	NO	Yucelen and Kidnay, 1999[55]
3	DA	99.99	99.95	%	%	270	0.01	4.28-31.6	0.1	YES	YES	YES	NO	Yucelen and Kidnay, 1999[55]
3	DA	99.99	99.95	%	%	330	0.01	19.5-57.7	0.1	YES	YES	YES	NO	Yucelen and Kidnay, 1999[55]
4	DA	99.99	99.98	%	%	319.3	N/A	21.8-76.3	N/A	YES	YES	YES	YES	Hsu et al, 1985[23]

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures														
CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
4	DA	99.99	99.98	%	%	344.3	N/A	32.1-81.2	N/A	YES	YES	YES	YES	Hsu et al, 1985[23]
4	DA	99.99	99.98	%	%	377.6	N/A	28.8-75.8	N/A	YES	YES	YES	YES	Hsu et al, 1985[23]
4	SA	99.9	99.9	%	%	368.1	0.1	13.7-78	0.1	YES	YES	YES	NO	Leu and Robinson, 1987[24]
4	SA	99.9	99.9	%	%	393.1	0.1	21.8-66.8	0.1	YES	YES	YES	NO	Leu and Robinson, 1987[24]
4	SA	99.9	99.9	%	%	418.1	0.1	32.7-46.4	0.1	YES	YES	YES	NO	Leu and Robinson, 1987[24]
4	DA	N/A	N/A	N/A	N/A	310.9	0.05	3.6-74	0.07	YES	YES	NO	YES	Shibata and Sandler, 1989[56]
4	DA	N/A	N/A	N/A	N/A	344.3	0.05	8.3-80.6	0.07	YES	YES	NO	YES	Shibata and Sandler, 1989[56]
4	DA	N/A	N/A	N/A	N/A	410.9	0.05	30.1-51.6	0.07	YES	YES	NO	YES	Shibata and Sandler, 1989[56]
4	DA	99.99	99.94	%	%	309.1	0.01	3.4-24.6	0.03	YES	YES	YES	NO	Weber et al., 1989[57]
4	DA	99.99	99.94	%	%	344.3	0.01	8.3-72.9	0.03	YES	YES	YES	NO	Weber et al., 1989[57]
4	DA	99.99	99.94	%	%	369.3	0.01	14.2-78.5	0.03	YES	YES	YES	NO	Weber et al., 1989[57]
4	DA	99.99	99.94	%	%	394.3	0.01	22.5-65.6	0.03	YES	YES	YES	NO	Weber et al., 1989[57]
4	SA	> 99.99	> 99.95	%	%	310.9	0.1	11-70.8	0.3	YES	YES	YES	NO	Traub and Stephan, 1990[58]
4	SA	0.999	0.999	Mole fr.	Mole fr.	310.9	0.06	4.4-65.5	0.01	YES	YES	YES	YES	Olds et al., 1949[22]
4	SA	0.999	0.999	Mole fr.	Mole fr.	344.3	0.06	10.3-79.7	0.01	YES	YES	YES	YES	Olds et al., 1949[22]
4	SA	0.999	0.999	Mole fr.	Mole fr.	377.6	0.06	21.4-75.2	0.01	YES	YES	YES	YES	Olds et al., 1949[22]
4	SA	0.999	0.999	Mole fr.	Mole fr.	410.9	0.06	42.6-51.9	0.01	YES	YES	YES	YES	Olds et al., 1949[22]
4	DA	99.9	99.9	%	%	250	0.02	0.39-16.3	0.0035	YES	YES	YES	YES	Brown et al, 1989[59]
4	DA	99.9	99.9	%	%	270	0.02	0.9-27.7	0.0035	YES	YES	YES	YES	Brown et al, 1989[59]
5	SA	99.9	99.4	%	%	408.1	0.1	11.8-88.8	0.1	YES	YES	YES	NO	Leu and Robinson, 1987[26]
5	SA	99.9	99.4	%	%	438.1	0.1	20.2-69.5	0.1	YES	YES	YES	NO	Leu and Robinson, 1987[26]
5	SA	99.9	99.4	%	%	463.1	0.1	29.5-46.3	0.1	YES	YES	YES	NO	Leu and Robinson, 1987[26]
5	DA	99.99	> 99	%	%	252.7	0.01	1.6-17.7	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	273.4	0.01	2.6-32.	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	294.1	0.01	1.7-51.7	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	311.6	0.01	4.1-70.4	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	344.3	0.01	4.6-89.8	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	377.7	0.01	7.8-94.9	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	394.3	0.01	9.7-90.2	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	423.5	0.01	19.7-76.3	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures

CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
5	DA	99.99	> 99	%	%	442.5	0.01	26.2-60.6	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	DA	99.99	> 99	%	%	458.5	0.01	31.2-43.5	0.07	YES	YES	YES	NO	Cheng et al, 1989[60]
5	SA	99.99	99.6	%	%	310.1	0.03	5.6-72.1	0.3	YES	YES	YES	NO	Tochigi et al, 1998[61]
5	SA	99.99	99.6	%	%	333.1	0.03	5.9-86.4	0.3	YES	YES	YES	NO	Tochigi et al, 1998[61]
5	SA	99.99	99.6	%	%	363.1	0.03	9.3-95.4	0.3	YES	YES	YES	NO	Tochigi et al, 1998[61]
6	SA	99.99	> 99.99	%	%	298.1	N/A	4.4-51.4	0.01	YES	YES	NO	NO	Ohgaki and Katayama, 1976[62]
6	SA	99.99	> 99.99	%	%	313.1	N/A	6.2-75.6	0.01	YES	YES	NO	NO	Ohgaki and Katayama, 1976[62]
6	SS	> 99.9	> 99.8	%	%	273.1	0.01	10.7-31.5	0.001	YES	NO	YES	NO	Kaminishi et al, 1987[54]
6	SS	> 99.9	> 99.8	%	%	283.1	0.01	12.4-40.1	0.001	YES	NO	YES	NO	Kaminishi et al, 1987[54]
6	SS	> 99.9	> 99.8	%	%	298.1	0.01	15.3-55.4	0.001	YES	NO	YES	NO	Kaminishi et al, 1987[54]
6	SS	> 99.9	> 99.8	%	%	303.1	0.01	16.2-61.1	0.001	YES	NO	YES	NO	Kaminishi et al, 1987[54]
6	SA	99.68	> 99.9	%	%	303.1	0.002	20.2-67.5	0.02	YES	YES	YES	NO	Wagner and Wichtierle, 1987[63]
6	SA	99.68	> 99.9	%	%	313.1	0.002	20.8-78.7	0.02	YES	YES	YES	NO	Wagner and Wichtierle, 1987[63]
6	SA	99.68	> 99.9	%	%	323.1	0.002	18.6-84.9	0.02	YES	YES	YES	NO	Wagner and Wichtierle, 1987[63]
6	SA	99.99	99.0	Mole %	Mole %	313.1	0.05	7.8-74.8	0.14	YES	YES	N/A	NO	Li et al, 1981[64]
6	SA	99.99	99.0	Mole %	Mole %	353.1	0.05	8.6-106.6	0.14	YES	YES	N/A	NO	Li et al, 1981[64]
6	SA	99.99	99.0	Mole %	Mole %	393.1	0.05	9.0-116.0	0.14	YES	YES	N/A	NO	Li et al, 1981[64]
7	SA	> 99.9	> 99.9	Mole %	Mole %	310.6	0.18	1.9-75.6	0.21	YES	YES	YES	YES	Kalra et al., 1978[29]
7	SA	> 99.9	> 99.9	Mole %	Mole %	352.6	0.18	4.2-116.1	0.21	YES	YES	YES	YES	Kalra et al., 1978[29]
7	SA	> 99.9	> 99.9	Mole %	Mole %	394.3	0.18	11.3-133.1	0.21	YES	YES	YES	YES	Kalra et al., 1978[29]
7	SA	> 99.9	> 99.9	Mole %	Mole %	477.2	0.18	17.5-99.2	0.21	YES	YES	YES	YES	Kalra et al., 1978[29]
7	DA	99.9	>99.0	Vol %	Vol%	394.0	0.1	50-130	0.2	YES	YES	YES	NO	Inomata et al, 1986 [65]
8	DA	99.99	99	%	%	313.2	0.04	15-75	0.4	YES	YES	YES	NO	Weng and Lee 1992[66]
8	DA	99.99	99	%	%	328.2	0.04	20-95	0.4	YES	YES	YES	NO	Weng and Lee 1992[66]
8	DA	99.99	99	%	%	348.2	0.04	20-113.5	0.4	YES	YES	YES	NO	Weng and Lee 1992[66]
8	SA	99.995	99	%	Mole %	322.4	0.03	20.1-85.3	0.08	YES	YES	NO	NO	Jiménez-Gallegos 2006[67]
8	SA	99.995	99	%	Mole %	348.3	0.03	20.7-114.6	0.08	YES	YES	NO	NO	Jiménez-Gallegos 2006[67]
8	SA	99.995	99	%	Mole %	372.5	0.03	31.5-137.7	0.08	YES	YES	NO	NO	Jiménez-Gallegos 2006[67]
8	SA	99.995	> 99	Mole %	%	322.4	0.03	20.1-85.3	0.06	YES	YES	YES	NO	Jiménez-Gallegos et al., 2006[67]
8	SA	99.995	> 99	Mole %	%	348.2	0.03	20.7-114.6	0.06	YES	YES	YES	NO	Jiménez-Gallegos et al., 2006[67]
8	SA	99.995	> 99	Mole %	%	372.5	0.03	31.5-137.7	0.06	YES	YES	YES	NO	Jiménez-Gallegos et al., 2006[67]
9	DA	99.999	99.99	%	%	343.2	0.1	37.3-118.0	0.07	YES	YES	YES	NO	Jennings and Schucker, 1996 [68]
9	DA	99.999	99	%	%	343.2	0.3	37.3-118	0.2	YES	YES	NO	NO	Jennings and Schucker 1996[68]
10	DA	99.999	99.99	%	%	344.2	0.1	43.3-118.5	0.07	YES	YES	YES	NO	Jennings and Schucker, 1996[68]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	277.6	2	3.45-39.1	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	310.9	0.09	6.9-80.0	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures														
CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	344.3	0.09	13.8-128.2	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	377.6	0.09	13.8-164.9	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	410.9	0.09	13.8-185.6	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	444.3	0.09	13.8-188.4	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	477.6	0.09	13.8-178.4	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	SA	0.9995	0.9934	Mole fr.	Mole fr.	510.9	0.09	13.8-153.3	0.01	YES	YES	YES	YES*	Reamer and Sage, 1963[31]
10	DA	99.9	99.9	%	%	313.0	0.02	10.0-77.9	0.1	NO	YES	YES	NO	Chylinski and Gregorowicz 1998[69]
10	DA	> 99.9	> 99	Volume %	%	311.0	0.1	45.5-68.6	0.18	YES	YES	NO	NO	Iwai et al., 1994[70]
10	DA	> 99.9	> 99	Volume %	%	344.3	0.1	55.1-118.5	0.18	YES	YES	NO	NO	Iwai et al., 1994[70]
10	DA	99.9	> 99.0	Volume %	Volume %	411.2	0.1	50-180	0.2	YES	YES	YES	NO	Inomata et al, 1986[65]
10	DA	99.99	99.0	%	%	344.3	N/A	63.85-127.4	0.14	YES	YES	NO	YES	Nagarajan and Robinson , 1986[32]
10	DA	99.99	99.0	%	%	377.6	N/A	103.4-164.9	0.14	YES	YES	NO	YES	Nagarajan and Robinson , 1986[32]
10	DA	> 99.99	> 99.0	%	%	462.5	0.11	19.6-51.3	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
10	DA	> 99.99	> 99.0	%	%	476.9	0.11	14.4-50.7	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
10	DA	> 99.99	> 99.0	%	%	542.9	0.11	19.6-51.7	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
10	DA	> 99.99	> 99.0	%	%	583.6	0.25	20.0-51.0	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
10	DA	99.99	99.8	%	%	310.9	0.02	27.8-80.4	0.08	YES	NO	NO	YES	Han et al, 1992[72]
10	DA	99.98	99.0	%	%	313.1	0.01	14.3-78.3	0.16	YES	YES	YES	YES	Adams et al, 1988[73]
10	DA	99.9	99.9	%	%	313	0.02	10-77.9	0.1	NO	YES	YES	NO	Chylinski and Gregorowicz 1998[69]
10	DA	99.999	99	%	%	343.2	0.3	43.3-118.5	0.2	YES	YES	NO	NO	Jennings and Schucker 1996[68]
10	DA	99.5	99	%	%	344.3	0.06	8.9-127	0.14	YES	YES	YES	YES	Shaver et al. 2001[74]
10	SA	99.99	99	%	%	344.8	0.1	2.6-11	0.05	YES	NO	YES	YES	Tsuji et al. 2004[75]
10	SA	99.995	> 99	Mole %	%	319.1	0.03	34.8-89	0.06	YES	YES	YES	NO	Jiménez-Gallegos et al., 2006[67]
10	SA	99.995	> 99	Mole %	%	344.7	0.03	45.8-126.6	0.06	YES	YES	YES	NO	Jiménez-Gallegos et al., 2006[67]
10	SA	99.995	> 99	Mole %	%	372.9	0.03	32.4-160.6	0.06	YES	YES	YES	NO	Jiménez-Gallegos et al., 2006[67]
12	SS	N/A	> 99	-	%	313.2	N/A	79.5-85.7	< 1	YES	YES	NO	NO	Nieuwoudt and du Rand 2002[76]
12	SS	N/A	> 99	-	%	323.2	N/A	94.4-102.9	< 1	YES	YES	NO	NO	Nieuwoudt and du Rand 2002[76]
12	SS	N/A	> 99	-	%	333.2	N/A	110.4-123.3	< 1	YES	YES	NO	NO	Nieuwoudt and du Rand 2002[76]
12	SS	N/A	> 99	-	%	343.2	N/A	120.0-143	< 1	YES	YES	NO	NO	Nieuwoudt and du Rand 2002[76]
12	SA	99.995	99.9	%	%	318.1	0.5	9.5-45.28	0.007	YES	YES	YES	NO	Gardeler et al, 2002[77]
14	DA	99.99	99	%	%	344.3	N/A	69-163.8	N/A	YES	YES	NO	YES	Gasem et al. 1989[78]
15	SA	99.999	99	%	%	313.2	N/A	17.0-64.1	N/A	YES	NO	YES	YES	Tanaka and Yamaki, 1993[79]
15	SA	0.997	0.99	Mass fr.	Mass fr.	298.1	0.1	5.9-127.8	1.25	YES	YES	YES	NO	Secuianu et al., 2007[45]
15	SA	0.997	0.99	Mass fr.	Mass fr.	316.1	0.1	5.1-105.3	1.25	YES	YES	YES	NO	Secuianu et al., 2007[45]
15	SA	0.997	0.99	Mass fr.	Mass fr.	293.1	0.1	7.6-141.2	0.1	YES	YES	YES	NO	Secuianu et al., 2010[46]
15	SA	0.997	0.99	Mass fr.	Mass fr.	303.1	0.1	7.2-100.3	0.1	YES	YES	YES	NO	Secuianu et al., 2010[46]

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures

CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
15	SA	0.997	0.99	Mass fr.	Mass fr.	313.1	0.1	5.6-110.3	0.1	YES	YES	YES	NO	Secuianu et al., 2010[46]
15	SA	0.997	0.99	Mass fr.	Mass fr.	333.1	0.1	6.8-139.4	0.1	YES	YES	YES	NO	Secuianu et al., 2010[46]
15	SA	0.997	0.99	Mass fr.	Mass fr.	353.1	0.1	9.3-175	0.1	YES	YES	YES	NO	Secuianu et al., 2010[46]
16	SS	N/A	99	-	%	313.2	N/A	85.5-165.1	< 1	YES	YES	NO	NO	Nieuwoudt and du Rand 2002[76]
16	SS	N/A	99	-	%	323.2	N/A	101.5-168.4	< 1	YES	YES	NO	NO	Nieuwoudt and du Rand 2002[76]
16	DA	N/A	N/A	N/A	N/A	343.1	N/A	6.9-258.7	N/A	YES	YES	NO	YES	Charoensombut-Amon et al, 1986 [38]
16	DA	N/A	N/A	N/A	N/A	333.1	N/A	6.9-218.6	N/A	YES	YES	NO	YES	Charoensombut-Amon et al, 1986 [38]
16	DA	N/A	N/A	N/A	N/A	323.1	N/A	6.9-144.3	N/A	YES	YES	NO	YES	Charoensombut-Amon et al, 1986 [38]
16	DA	N/A	N/A	N/A	N/A	313.1	N/A	6.9-122.9	N/A	YES	YES	NO	YES	Charoensombut-Amon et al, 1986 [38]
16	DA	N/A	N/A	N/A	N/A	308.1	N/A	6.9-144.3	N/A	YES	YES	NO	NO	Charoensombut-Amon et al, 1986 [38]
16	SA	N/A	> 99.5	N/A	%	393.2	0.2	101.0-256.0	1.0	YES	YES	YES	NO	Spee and Schneider, 1991[39]
16	SA	N/A	> 99.0	N/A	%	353.2	0.1	100.0-200.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]
16	DA	> 99.99	> 99.0	%	%	463.0	0.11	20.1-51.7	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
16	DA	> 99.99	> 99.0	%	%	542.8	0.11	20.1-50.8	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
16	DA	> 99.99	> 99.0	%	%	623.5	0.11	20.2-50.6	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
16	DA	> 99.99	> 99.0	%	%	663.7	0.11	20.9-50.6	0.03	YES	YES	NO	NO	Sebastian et al, 1980[71]
16	SA	99.5	> 99.0	%	%	473.1	1.0	101.0-257.0	0.5	YES	YES	YES	NO	Brunner et al, 1994[80]
16	SA	99.5	> 99.0	%	%	573.1	1.0	101.0-213.0	0.5	YES	YES	YES	NO	Brunner et al, 1994[80]
16	SA	N/A	99	N/A	%	333.2	0.2	5-290	0.1	YES	YES	YES	NO	Holscher et al 1989[81]
16	SA	N/A	99	N/A	%	393.2	0.2	5-290	0.1	YES	YES	YES	NO	Holscher et al 1989[81]
16	DA	99.5	99	%	%	305.2	0.45	100-130	0.04	NO	YES	YES	NO	Rose et al. 2000[82]
16	DA	99.5	99	%	%	311.2	0.45	70-180	0.04	NO	YES	YES	NO	Rose et al. 2000[82]
16	SA	99.999	98	%	%	313.2	N/A	17.2-64.7	N/A	YES	NO	YES	YES	Tanaka et al. 1993[79]
16	DA	99.9	99	Mole %	Mole %	314.2	0.01	80.6-136.1	0.07	YES	YES	NO	NO	D'Souza et al. 1988[83]
16	DA	99.9	99	Mole %	Mole %	333.2	0.01	76.9-148.7	0.07	YES	YES	NO	NO	D'Souza et al. 1988[83]
16	DA	99.9	99	Mole %	Mole %	353.2	0.01	112.1-161.2	0.07	YES	YES	NO	NO	D'Souza et al. 1988[83]
18	DA	99.998	98	Mole %	Mole %	310	0.05	100-200	0.002	NO	YES	YES	NO	Eustaquio-Rincón, Trejo 2001[84]
18	DA	99.998	98	Mole %	Mole %	313	0.05	100-200	0.002	NO	YES	YES	NO	Eustaquio-Rincón, Trejo 2001[84]
18	DA	99.998	98	Mole %	Mole %	333	0.05	100-200	0.002	NO	YES	YES	NO	Eustaquio-Rincón, Trejo 2001[84]
18	DA	99.998	98	Mole %	Mole %	353	0.05	100-200	0.002	NO	YES	YES	NO	Eustaquio-Rincón, Trejo 2001[84]
18	SA	N/A	N/A	N/A	N/A	353.2	0.1	90-260	1	YES	YES	NO	NO	Pöhler et al. 1996[36]
18	SA	N/A	N/A	N/A	N/A	373.2	0.1	90-270	1	YES	YES	NO	NO	Pöhler et al. 1996[36]
18	SA	N/A	N/A	N/A	N/A	393.2	0.1	90-270	1	YES	YES	NO	NO	Pöhler et al. 1996[36]
19	SAV	N/A	> 99.0	N/A	%	353.2	0.1	100.0-275.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]
19	SAV	N/A	> 99.0	N/A	%	393.2	0.1	100.0-303.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]
19	SA	99.99	99.9	%	%	313	0.1	2.9-86.4	0.07	YES	NO	YES	YES	Elbaccouch et al. 2003[85]

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures														
CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
19	SA	99.99	99	%	%	313.1	0.02	11.2-79.1	0.07	YES	NO	YES	YES	Fall et al. 1985[17]
19	SA	99.99	99	%	%	333.1	0.02	9.4-79.6	0.07	YES	NO	YES	YES	Fall et al. 1985[17]
20	SS	N/A	99	N/A	%	315.9	N/A	77.4-249.6	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
20	SS	N/A	99	N/A	%	320.6	N/A	86.4-223.1	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
20	SS	N/A	99	N/A	%	329.6	N/A	104.3-258.9	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
20	SS	N/A	99	N/A	%	339.1	N/A	104.3-303.5	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
20	SS	N/A	99	N/A	%	348.3	N/A	115.0-293.2	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
20	SA	N/A	> 99.0	N/A	%	353.2	0.1	100.0-302.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]
20	SA	N/A	> 99.0	N/A	%	393.5	0.1	100.0-320.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]
20	SS	99.5	99.0	%	%	323.2	0.05	59.1-147.8	0.8	YES	NO	YES	NO	Sato et al, 1998 [86]
20	SS	99.5	99.0	%	%	373.2	0.05	87.9-307.8	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
20	SS	99.5	99.0	%	%	423.2	0.05	109.2-335.7	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
20	SS	99.5	99.0	%	%	473.2	0.05	122.1-344.1	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
20	SA	99.99	99	Mole %	Mole %	310.2	0.05	5.1-76.0	0.07	YES	NO	YES	YES	Huie et al. 1973[47]
20	SA	99.99	99	Mole %	Mole %	323.2	0.05	5.1-76.0	0.07	YES	NO	YES	YES	Huie et al. 1973[47]
20	SA	99.99	99	Mole %	Mole %	348.2	0.05	5.1-76.0	0.07	YES	NO	YES	YES	Huie et al. 1973[47]
20	SA	99.99	99	Mole %	Mole %	373.2	0.05	5.1-76.0	0.07	YES	NO	YES	YES	Huie et al. 1973[47]
20	DA	99.5	99	%	%	322.9	N/A	5.4-24.8	N/A	YES	NO	YES	NO	Feng and Mather 1992[87]
21	SA	99.99	99	%	%	318.1	0.02	11.5-77.6	0.07	YES	NO	YES	YES	Fall et al. 1985[17]
21	SA	99.99	99	%	%	338.1	0.02	9.3-78.2	0.07	YES	NO	YES	YES	Fall et al. 1985[17]
22	SS	99.5	98.0	%	%	323.2	0.05	63.7-123.8	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
22	SS	99.5	98.0	%	%	373.2	0.05	96.1-352.1	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
22	SS	99.5	98.0	%	%	423.2	0.05	121.7-366.4	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
22	SS	99.5	98.0	%	%	473.2	0.05	135.8-373.0	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
22	SA	99.99	99	%	%	323.2	0.01	11.1-71.8	0.07	YES	NO	YES	YES	Fall and Luks 1984[88]
22	SA	99.99	99	%	%	348.2	0.01	11.1-71.6	0.07	YES	NO	YES	YES	Fall and Luks 1984[88]
22	SA	99.99	99	%	%	373.2	0.01	9.6-65.6	0.07	YES	NO	YES	YES	Fall and Luks 1984[88]
22	SA	N/A	99	N/A	%	343.2	0.1	100-430	1	YES	YES	NO	NO	Pöhler and Schneider 1995[42]
22	SA	N/A	99	N/A	%	353.2	0.1	100-390	1	YES	YES	NO	NO	Pöhler and Schneider 1995[42]
22	SA	N/A	99	N/A	%	373.2	0.1	100-370	1	YES	YES	NO	NO	Pöhler and Schneider 1995[42]
22	SA	N/A	99	N/A	N/A	393.2	0.1	100-370	1	YES	YES	NO	NO	Pöhler and Schneider 1995[42]
24	SS	N/A	99	N/A	%	329.7	N/A	154.8-281.4	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
24	SS	N/A	99	N/A	%	339.2	N/A	166.0-283.7	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
24	SS	N/A	99	N/A	%	348.1	N/A	172.4-279.8	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
24	SS	N/A	99	N/A	%	357.1	N/A	185.0-301.6	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
24	SA	N/A	> 99.0	N/A	%	353.2	0.1	100.0-475.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]

Table A4. Isothermal Data Sets of Experimental Fluid-Fluid Equilibria (Pxy) available for CO_2 + n-Alkane binary mixtures														
CN	Method	Purity		Units for Purity		T (K)	Absolute T Uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
24	SA	N/A	> 99.0	N/A	%	393.2	0.1	98.0-300.0	1.0	YES	YES	NO	NO	Kordikowski and Schneider, 1993[41]
24	SS	99.5	99.0	%	%	373.2	0.05	94.3-389.4	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
24	SS	99.5	99.0	%	%	423.2	0.05	122.8-348.8	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
24	SS	99.5	99.0	%	%	473.2	0.05	140.9-311.3	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
24	DA	99	99	%	%	373.1	0.2	10.1-50.6	0.002	YES	YES	NO	NO	Tsai and Yau 1990[89]
24	DA	99	99	%	%	473.1	0.2	10.1-50.6	0.002	YES	YES	NO	NO	Tsai and Yau 1990[89]
24	DA	99	99	%	%	573.1	0.2	10.1-50.6	0.002	YES	YES	NO	NO	Tsai and Yau 1990[89]
28	SS	99.5	98.0	%	%	373.2	0.05	89.7-222.8	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
28	SS	99.5	98.0	%	%	423.2	0.05	115.0-408.5	0.8	YES	NO	YES	NO	Sato et al, 1998[86]
28	SS	N/A	≥ 98	N/A	%	338.5	N/A	140.1-278.7	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
28	SS	N/A	≥ 98	N/A	%	348.0	N/A	148.4-289.9	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
28	SS	N/A	≥ 98	N/A	%	357.0	N/A	162.3-284.4	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
28	SS	N/A	≥ 98	N/A	%	366.4	N/A	172.3-289.5	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
28	DA	99.99	99	%	%	308	0.1	80-220	0.5	NO	YES	YES	YES	Reverchon et al. 1993[90]
28	DA	99.99	99	%	%	313	0.1	90-275	0.5	NO	YES	YES	YES	Reverchon et al. 1993[90]
28	DA	99.99	99	%	%	318	0.1	105-250	0.5	NO	YES	YES	YES	Reverchon et al. 1993[90]
28	DA	99.8	99	%	%	373.4	0.1	10.6-50.5	0.05	YES	NO	NO	NO	Huang et al. 1988[91]
28	DA	99.8	99	%	%	473.5	0.1	10-50.6	0.05	YES	NO	NO	NO	Huang et al. 1988[91]
28	DA	99.8	99	%	%	573.4	0.1	9.9-50.5	0.05	YES	YES	NO	NO	Huang et al. 1988[91]
30	DA	99.99	99	%	%	308	0.1	90-250	0.5	NO	YES	YES	YES	Reverchon et al. 1993[90]
30	DA	99.99	99	%	%	318	0.1	105-250	0.5	NO	YES	YES	YES	Reverchon et al. 1993[90]
32	SAV	N/A	> 97.0	N/A	%	393.2	0.2	100-630	1.0	YES	YES	YES	NO	Spee and Schneider, 1991[39]
32	SA	99.99	99	%	%	348	0.01	9.5-72.1	0.07	YES	NO	YES	YES	Fall and Luks 1984[88]
32	SA	99.99	99	%	%	373	0.01	11.6-72.2	0.07	YES	NO	YES	YES	Fall and Luks 1984[88]
32	SA	99.99	99	%	%	398	0.01	9.5-72.3	0.07	YES	NO	YES	YES	Fall and Luks 1984[88]
32	DA	99	99	%	%	373.1	0.2	10.1-50.6	0.11	YES	YES	NO	NO	Tsai and Yau 1990[89]
32	DA	99	99	%	%	473.1	0.2	10.1-50.6	0.1	YES	YES	NO	NO	Tsai and Yau 1990[89]
32	DA	99	99	%	%	573.1	0.2	10.1-50.6	0.1	YES	YES	NO	NO	Tsai and Yau 1990[89]
36	SS	N/A	98	N/A	%	344.0	N/A	201.0-289.6	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
36	SS	N/A	98	N/A	%	348.5	N/A	201.4-297.5	< 1	YEZ	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
36	SS	N/A	98	N/A	%	357.0	N/A	205.3-291.3	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]
36	SS	N/A	98	N/A	%	366.9	N/A	211.5-289.7	< 1	YES	YES	NO	YES	Nieuwoudt and du Rand 2002[76]

* Density informed for some points. SA: Static analytic. SAV: Static analytic visual. SS: Static synthetic. DA: Dynamic Analytic. DS: Dynamic synthetic. Mole fr.: Mole fraction. Mass fr.: Mass fraction

Table A5. Isobaric Data Sets of Experimental Fluid-Fluid Equilibria (Txy) available for CO_2 + n-Alkane binary mixtures														
CN	Method	Units for Purity		Purity		P (bar)	Absolute P uncertainty (bar)	T range (K)	Absolute T uncertainty (K)	Alk. richer phase	CO_2 richer phase	Composition uncertainty informed	Density informed	Reference
		CO_2	Alk	CO_2	Alk									
16	Static analytic	N/A	> 99.5	N/A	%	250.0	1.0	294.4-413.3	0.2	YES	YES	YES	NO	Spee and Schneider, 1991[39]
19	Static analytic	N/A	> 99.0	N/A	%	250	1.0	303.1-383.1	0.1	YES	YES	N/A	NO	Kordikowski and Schneider, 1993[41]
22	Analytic static	N/A	99	N/A	%	34.5	1	330-400	0.1	YES	YES	NO	NO	Pöhler and Schneider 1995[42]

Table A6. Isoplethic Data Sets of Experimental Fluid-Fluid Equilibria available for CO_2 + n-Alkane binary mixtures													
CN	Method	Purity		Units for Purity		Composition* (z_{CO2})	Absolute composition uncertainty (mole fraction)	T range (K)	Absolute T uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Density informed	Reference
		CO_2	Alk	CO_2	Alk								
2	SS	0.99995	0.9996	Mole fr.	Mole fr.	0.7397	N/A	292.7-285.1	0.11	61.59-51.56	0.11	YES	Goodwin and Moldover, 1997[92]
2	SS	0.99995	0.9996	Mole fr.	Mole fr.	0.4924	N/A	291.1-285.0	0.06	55.99-49.54	0.07	YES	Goodwin and Moldover, 1997[92]
2	SS	0.99995	0.9996	Mole fr.	Mole fr.	0.4924	N/A	287.3-283.7	0.13	52.03-46.7	0.15	YES	Goodwin and Moldover, 1997[92]
2	SS	0.99995	0.9996	Mole fr.	Mole fr.	0.2516	N/A	295.4-292.4	0.25	52.32-46.84	0.28	YES	Goodwin and Moldover, 1997[92]
3	SA	99.5	99.9	%	%	0.7873	0.003	290.4-311.7	0.3	33.7-69.1	0.3	NO	Poettmann and Katz, 1945[20]
3	SA	99.5	99.9	%	%	0.3599	0.001	295.6-347.6	0.3	12.5-66.7	0.3	NO	Poettmann and Katz, 1945[20]
3	SA	99.5	99.9	%	%	0.5065	0.002	299.9-337.0	0.3	19.0-68.0	0.3	NO	Poettmann and Katz, 1945[20]
3	SA	99.5	99.9	%	%	0.1519	0.0005	303.7-366.1	0.3	11.0-54.5	0.3	NO	Poettmann and Katz, 1945[20]
3	SA	99.5	99.9	%	%	0.9601	0.004	298.9-303.9	0.3	62.9-71.1	0.3	NO	Poettmann and Katz, 1945[20]
3	SA	99.5	99.9	%	%	0.9393	0.004	291.4-304.4	0.3	51.0-70.5	0.3	NO	Poettmann and Katz, 1945[20]
3	DA	99.995	99.99	%	%	0.076	0.001	213.2- 349.8	0.05	0.59-35.74	0.1 %	No	Acosta et al, 1984[93]
3	DA	99.995	99.99	%	%	0.181	0.001	210.9- 349.8	0.05	0.62-46.08	0.1 %	No	Acosta et al, 1984[93]
3	DA	99.995	99.99	%	%	0.244	0.001	216.5- 333.2	0.05	0.83-38.46	0.1 %	No	Acosta et al, 1984[93]
3	DA	99.995	99.99	%	%	0.349	0.001	216.5-333.2	0.05	0.76-51.21	0.1 %	No	Acosta et al, 1984[93]
3	DA	99.995	99.99	%	%	0.442	0.001	216.5- 327.6	0.05	0.76-58.59	0.1 %	No	Acosta et al, 1984[93]
4	SA	99.5	99.0	%	%	0.4551	0.002	305.5-388.8	0.3	6.6-74.6	0.3	NO	Poettmann and Katz, 1945[20]
4	SA	99.5	99.0	%	%	0.7102	0.003	306.1-355.1	0.3	9.6-81.6	0.3	NO	Poettmann and Katz, 1945[20]
4	SA	99.5	99.0	%	%	0.8609	0.003	306.8-324.8	0.3	30.7-77.5	0.3	NO	Poettmann and Katz, 1945[20]
4	SA	99.5	99.0	%	%	0.6073	0.003	299.9-374.8	0.3	6.1-78.8	0.3	NO	Poettmann and Katz, 1945[20]
4	SA	99.5	99.0	%	%	0.3761	0.002	300.8-397.1	0.3	4.5-66.1	0.3	NO	Poettmann and Katz, 1945[20]
4	SA	99.5	99.0	%	%	0.1393	0.005	301.7-416.4	0.3	2.9-50.0	0.3	NO	Poettmann and Katz, 1945[20]
5	SA	99.5	99.0	%	%	0.9511	0.004	295.3-315.1	0.3	31.0-79.8	0.3	NO	Poettmann and Katz, 1945[20]
5	SA	99.5	99.0	%	%	0.8967	0.004	299.8-338.9	0.3	19.0-88.5	0.3	NO	Poettmann and Katz, 1945[20]
5	SA	99.5	99.0	%	%	0.7938	0.003	299.8-383.1	0.3	5.2-99.4	0.3	NO	Poettmann and Katz, 1945[20]
5	SA	99.5	99.0	%	%	0.6687	0.003	299.7-411.4	0.3	1.6-95.3	0.3	NO	Poettmann and Katz, 1945[20]
5	SA	99.5	99.0	%	%	0.4553	0.002	299.9-399.9	0.3	1.4-64.4	0.3	NO	Poettmann and Katz, 1945[20]
6	SS	99.9	99.9	%	%	0.974	0.01461	310.7-314.4	0.1	77.8-81.0	0.6	NO	Choi and Yeo, 1998[27]
6	SS	99.9	99.9	%	%	0.922	0.01383	313.1-343.5	0.1	77.5-97.7	0.6	NO	Choi and Yeo, 1998[27]
6	SS	99.9	99.9	%	%	0.880	0.0132	314.3-358.2	0.1	71.8-106.7	0.6	NO	Choi and Yeo, 1998[27]
6	SS	99.9	99.9	%	%	0.871	0.013065	313.8-346.0	0.1	75.6-101.3	0.6	NO	Choi and Yeo, 1998[27]

Table A6. Isoplethic Data Sets of Experimental Fluid-Fluid Equilibria available for CO_2 + n-Alkane binary mixtures													
CN	Method	Purity		Units for Purity		Composition* (z_{CO2})	Absolute composition uncertainty (mole fraction)	T range (K)	Absolute T uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Density informed	Reference
		CO_2	Alk	CO_2	Alk								
6	SS	99.9	99.9	%	%	0.834	0.01251	313.7-358.1	0.1	75.3-103.7	0.6	NO	Choi and Yeo, 1998[27]
7	SS	99.9	99.9	%	%	0.958	0.01437	313.2-327.9	0.1	82.1-96.0	0.6	NO	Choi and Yeo, 1998[27]
7	SS	99.9	99.9	%	%	0.925	0.013875	328.7-338.3	0.1	96.0-104.3	0.6	NO	Choi and Yeo, 1998[27]
7	SS	99.9	99.9	%	%	0.904	0.01356	313.1-363.3	0.1	75.5-119.1	0.6	NO	Choi and Yeo, 1998[27]
7	SS	99.9	99.9	%	%	0.895	0.013425	322.9-368.1	0.1	83.6-121.5	0.6	NO	Choi and Yeo, 1998[27]
7	SS	99.9	99.9	%	%	0.885	0.013275	323.6-370.1	0.1	85.5-119.5	0.6	NO	Choi and Yeo, 1998[27]
7	SS	0.99995	0.99	Mole fr.	Mole fr.	0.2918	N/A	362.9-459.4	0.01	43.8-500.2	0.5	YES	Fenghour et al, 2001[94]
7	SS	0.99995	0.99	Mole fr.	Mole fr.	0.3888	N/A	321.1-404.7	0.01	40.4-554.8	0.5	YES	Fenghour et al, 2001[94]
7	SS	0.99995	0.99	Mole fr.	Mole fr.	0.4270	N/A	301.8-368.0	0.01	34.6-500.2	0.5	YES	Fenghour et al, 2001[94]
8	SS	99.9	99.9	%	%	0.958	0.01437	313.1-342.9	0.1	81.9-112.9	0.6	NO	Choi and Yeo, 1998[27]
8	SS	99.9	99.9	%	%	0.939	0.014085	313.1-338.4	0.1	76.9-106.8	0.6	NO	Choi and Yeo, 1998[27]
8	SS	99.9	99.9	%	%	0.916	0.01374	313.1-348.1	0.1	76.9-119.1	0.6	NO	Choi and Yeo, 1998[27]
8	SS	99.9	99.9	%	%	0.902	0.01353	313.1-363.3	0.1	77.8-133.7	0.6	NO	Choi and Yeo, 1998[27]
8	SS	99.9	99.9	%	%	0.878	0.01317	313.1-368.1	0.1	76.5-135.5	0.6	NO	Choi and Yeo, 1998[27]
8	SS	99.8	N/A	%	N/A	0.437	0.02	298.15-437.85	0.2	34.5-82.5	2	NO	Schneider et al., 1967[95]
8	SS	99.8	N/A	%	N/A	0.508	0.02	287.25-531.35	0.2	35-105	2	NO	Schneider et al., 1967[95]
8	SS	99.8	N/A	%	N/A	0.674	0.02	287.25-496.65	0.2	41-137.5	2	NO	Schneider et al., 1967[95]
8	SS	99.8	N/A	%	N/A	0.730	0.02	282.95-512.15	0.2	39-144	2	NO	Schneider et al., 1967[95]
8	SS	99.8	N/A	%	N/A	0.806	0.02	284.85-471.85	0.2	42.5-155.5	2	NO	Schneider et al., 1967[95]
8	SS	99.8	N/A	%	N/A	0.873	0.02	293.45-428.55	0.2	53.5-152	2	NO	Schneider et al., 1967[95]
8	SS	99.8	N/A	%	N/A	0.937	0.02	285.45-408.65	0.2	47.5-138.5	2	NO	Schneider et al., 1967[95]
9	SS	99.9	99.9	%	%	0.941	0.014115	323.4-351.5	0.1	95.3-127.8	0.6	NO	Choi and Yeo, 1998[27]
9	SS	99.9	99.9	%	%	0.925	0.013875	322.7-358.3	0.1	95.4-136.2	0.6	NO	Choi and Yeo, 1998[27]
9	SS	99.9	99.9	%	%	0.878	0.01317	322.7-395.7	0.1	91.4-153.4	0.6	NO	Choi and Yeo, 1998[27]
10	SA	> 99.8	> 99.0	%	%	0.675	N/A	344.2-444.4	0.1	97.4-163.8	0.3	NO	Chen et al, 1993[96]
11	SS	99.8	99.5	%	%	0.826	0.03	257.05-279.95	0.2	58-1337	10	NO	Schneider et al., 1967[95]
11	SS	99.8	99.5	%	%	0.862	0.03	257.65-280.45	0.2	39-1319	10	NO	Schneider et al., 1967[95]
11	SS	99.8	99.5	%	%	0.869	0.03	257.75-280.65	0.2	32-1316	10	NO	Schneider et al., 1967[95]
11	SS	99.8	99.5	%	%	0.880	0.03	257.65-280.25	0.2	29-1269	10	NO	Schneider et al., 1967[95]
11	SS	99.8	99.5	%	%	0.905	0.03	257.35-279.45	0.2	52-1252	10	NO	Schneider et al., 1967[95]
13	SS	99.8	99.8	%	%	0.772	0.04	268.65-315.15	0.2	83-1444	10	NO	Schneider et al., 1967[95]
13	SS	99.8	99.8	%	%	0.833	0.04	272.45-334.35	0.2	40-1463	10	NO	Schneider et al., 1967[95]

Table A6. Isoplethic Data Sets of Experimental Fluid-Fluid Equilibria available for CO_2 + n-Alkane binary mixtures													
CN	Method	Purity		Units for Purity		Composition* (z_{CO2})	Absolute composition uncertainty (mole fraction)	T range (K)	Absolute T uncertainty (K)	P range (bar)	Absolute P uncertainty (bar)	Density informed	Reference
		CO_2	Alk	CO_2	Alk								
13	SS	99.8	99.8	%	%	0.849	0.04	274.45-295.25	0.2	43-1435	10	NO	Schneider et al., 1967[95]
13	SS	99.8	99.8	%	%	0.887	0.04	275.05-328.85	0.2	48-1453	10	NO	Schneider et al., 1967[95]
13	SS	99.8	99.8	%	%	0.909	0.04	275.05-332.55	0.2	43-1435	10	NO	Schneider et al., 1967[95]
13	SS	99.8	99.8	%	%	0.948	0.04	273.05-326.25	0.2	39-1512	10	NO	Schneider et al., 1967[95]
16	SS	>99.9	>99	Mass %	Mass %	0.202	N/A	305.7-512.3	0.3	12.4-32.9	0.12	NO	Breman et al. 1994[97]
16	SS	99.8	99.8	%	%	0.846	0.05	293.55-367.55	0.2	144-1346	10	NO	Schneider et al., 1967[95]
16	SS	99.8	99.8	%	%	0.889	0.05	297.05-370.05	0.2	170-1268	10	NO	Schneider et al., 1967[95]
16	SS	99.8	99.8	%	%	0.913	0.05	297.95-338.95	0.2	168-1463	10	NO	Schneider et al., 1967[95]
16	SS	99.8	99.8	%	%	0.917	0.05	297.85-335.65	0.2	175-1464	10	NO	Schneider et al., 1967[95]
16	SS	99.8	99.8	%	%	0.928	0.05	296.95-358.05	0.2	174-1266	10	NO	Schneider et al., 1967[95]
16	SS	99.8	99.8	%	%	0.960	0.05	293.15-356.95	0.2	151-1473	10	NO	Schneider et al., 1967[95]
28	SS	>99.9	>97	Mass %	Mass %	0.611	N/A	397.9-519.4	0.3	18.7-25	0.12	NO	Breman et al. 1994[97]

SA: Static analytic. SS: Static synthetic. DA: Dynamic Analytic. DS: Dynamic synthetic. Mole fr.: Mole fraction. Mass fr.: Mass fraction.

* In mole fraction of CO_2.

APPENDIX B: Other tables to which the main text of the article refers

Table B1. Pure compound parameters* for the RK-PR EOS (This work)

Compound	Compound ID	a_c (bar*L^2/mol^2)	b (L/mol)	δ_1	K
Carbon Dioxide	CO_2	3.9809	0.026440	2.509688	2.04173
METHANE	C1	2.3570	0.029392	1.244398	1.44042
ETHANE	C2	5.8024	0.043368	1.583623	1.75359
PROPANE	C3	9.8513	0.059399	1.764917	1.93370
n-BUTANE	C4	14.6687	0.075559	1.904346	2.09604
n-PENTANE	C5	20.3887	0.092397	2.118901	2.24755
n-HEXANE	C6	26.7882	0.109456	2.313637	2.39124
n-HEPTANE	C7	33.7503	0.126979	2.465532	2.53450
n-OCTANE	C8	41.5046	0.145042	2.633607	2.67573
n-NONANE	C9	49.6476	0.163111	2.763240	2.80022
n-DECANE	C10	58.4693	0.182177	2.875941	2.94303
n-UNDECANE	C11	68.2601	0.201038	3.046729	3.02959
n-DODECANE	C12	77.8716	0.220190	3.134497	3.16265
n-TRIDECANE	C13	89.5311	0.241046	3.316538	3.25130
n-Tetradecane	C14	102.0200	0.259970	3.541600	3.27688
n-Pentadecane	C15	113.3595	0.279939	3.620746	3.39568
n-Hexadecane	C16	125.5040	0.299846	3.717471	3.46921
n-Heptadecane	C17	135.7470	0.319486	3.694975	3.63771
n-Octadecane	C18	148.1487	0.339557	3.788625	3.74081
n-Nonadecane	C19	160.5037	0.359993	3.845494	3.84899
n-Eicosane	C20	171.8450	0.380561	3.842323	4.01140
n-Heneicosane	C21	184.4799	0.402123	3.865807	4.10806
n-Docosane	C22	198.9758	0.420783	4.019107	4.15471
n-TRICOSANE	C23	210.9991	0.444393	3.959352	4.32337
n-TETRACOSANE	C24	223.9849	0.467426	3.952760	4.45024
n-PENTACOSANE	C25	236.1051	0.485341	3.995209	4.53313
n-HEXACOSANE	C26	250.9000	0.510475	4.009181	4.66289
n-HEPTACOSANE	C27	262.6818	0.531827	3.979683	4.82872
n-OCTACOSANE	C28	278.8538	0.549011	4.150393	4.84541
n-NONACOSANE	C29	291.5623	0.567363	4.187813	4.90774
n-TRIACONTANE	C30	305.8439	0.588238	4.225805	5.00535
n-DOTRIACONTANE	C32	333.3214	0.640966	4.119647	5.21010
n-HEXATRIACONTANE	C36	384.8594	0.720154	4.163653	5.56525

*They reproduce the experimental (DIPPR,[98]) critical temperature Tc, critical pressure Pc, acentric factor and liquid molar volume at 0.70 reduced temperature. Only for the case of CO_2, instead, the saturated liquid molar volume is reproduced at 270 K.

Table B2. Experimental fluid phase compositions under two-phase equilibrium conditions, at specified T and P, for some CO_2(1) + n-alkane(2) systems, considered in the objective function.

CN	*T* *(K)*	*P* *(bar)*	x_1	y_1	Reference
1	219.3	41.37	0.782	0.204	Mraw et al, 1978 [18] (Interpolated point)
1	270.0	65.37	0.8521	0.6298	Webster and Kidnay, 2001 [48]
2	230.0	9.48	0.1548	0.3225	Wei et al, 1995 [51]
2	230.0	10.23	0.9326	0.8389	Wei et al, 1995 [51]
2	270.0	31.51	0.2907	0.4116	Wei et al, 1995 [51]
3	270.0	17.0	0.3107	0.7658	Yucelen and Kidnay, 1999 [55]
4	344.3	55.3	0.4180	0.7770	Hsu et al., 1985 [23]
6	313.15	40.52	0.4446	0.9748	Wagner and Wichterle, 1987 [63]
10	344.3	85.5	0.5936	0.9968	Iwai et al., 1994 [70]
10	377.6	137.9	0.7173	0.9789	Reamer and Sage, 1963 [31]
16	393.2	101	0.4968	0.9982	Spee and Schneider, 1991 [39]
16	393.2	200	0.7473	0.9909	Spee and Schneider, 1991 [39]
16	573.2	101	0.428	0.962	Brunner et al., 1994 [80]
16	573.2	201	0.700	0.938	Brunner et al., 1994 [80]
20	353.2	250	0.78534	0.97959	Kordikowski and Schneider, 1993
20	393.2	250	0.70577	0.99422	Kordikowski and Schneider, 1993
22	373.2	199.55	0.76603	0.9963	Pohler and Schneider, 1995 [42]
22	373.2	299.47	0.86965	0.9862	Pohler and Schneider, 1995 [42]
24	573.2	50.66	0.2533	0.9961	Tsai and Yau, 1990 [89]
28	573.5	50.57	0.270	0.9983	Huang et al., 1988 [91]
32	573.2	50.66	0.2839	0.9996	Tsai and Yau, 1990 [89]

CN: Carbon number of the n-alkane. x_1 and y_1 are mole fractions of CO_2 in the heavy and light phases respectively.

Table B3. Experimental fluid-fluid saturation pressures at given temperature and liquid phase composition for some CO_2(1) + n-alkane(2) systems, considered in the objective function.

CN	*T* *(K)*	x_1	*P* *(bar)*	Reference
13	338.6	0.80	125	Enick et al., 19851985 [33]
13	377.4	0.80	178	Enick et al., 1985 1985 [33]
16	313.2	0.227	21.33	Tanaka et al., 1993 [79]
20	310.2	0.253	20.265	Huie et al., 1973 [47]
20	373.2	0.147	20.265	Huie et al., 1973 [47]
24	373.2	0.172	20.265	Tsai and Yau, 1990 [89]
24	573.2	0.1127	20.265	Tsai and Yau, 1990 [89]
28	373.4	0.184	20.40	Huang et al., 1988 [91]
28	573.5	0.122	20.33	Huang et al., 1988 [91]
28	373.2	0.7298	161.9	Sato et al., 1998 [86]
28	423.2	0.8972	408.5	Sato et al., 1998 [86]
32	348.2	0.2279	20.92	Fall and Luks, 1984 [88]
32	573.2	0.126	20.265	Tsai and Yau, 1990 [89]

CN: Carbon number of the n-alkane. x_1 is the mole fraction of CO_2 in the liquid phase.

Table B4. CMRs interaction parameters optimized for some CO_2(1) + n-alkane(2) systems individually

		SYSTEM						
		CO_2 + C1	CO_2 + C2	CO_2 + C8	CO_2 + C10	CO_2 + C13	CO_2 + C16	CO_2 + C20
Interaction Parameters	k'_{112}	0.02070	0.14971	0.20995	0.18520	0.22924	0.25047	0.27139
	k'_{122}	0.10795	0.25751	0.54902	0.52164	0.51408	0.48952	0.32785
	k^{∞}_{112}	0.00016	-0.04951	-0.18521	-0.22561	-0.22652	-0.25631	-0.31299
	k^{∞}_{122}	-0.02720	-0.14304	-0.59344	-0.64650	-0.67716	-0.74875	-0.83642
	l_{112}	-0.03829	-0.05656	0.00013	-0.01382	0.06752	0.09066	0.09198
	l_{122}	0.00732	0.00565	0.03503	0.02501	0.03952	0.05533	0.05224
	T^*_{112} /K	321.14	367.95	250.80	237.29	222.24	199.20	141.65
	T^*_{122} /K	1475.42	1857.5	980.64	720.28	799.39	981.09	1879.65
OF		0.1933	0.3416	0.652	1.683	0.715	1.759	1.842
OFcorr		-	-	0.958	1.860	0.972	2.040	3.220

OF is the objective function value corresponding to a single binary system optimization. Although some extra terms were used while fitting the "binary system-specific" interaction parameters for CO_2+C10, CO_2+C13, CO_2+C16 and CO_2+C20 (see section 3.3), the *OF* values reported here correspond only to the terms accounting for data in tables 2 to 5, B2 and B3, in order to be comparable to *OFcorr*.

OFcorr is the partial objective function value, corresponding to a given binary system, in the optimum found for the CO_2 + n-alkane series, covering simultaneously systems from C3 to C32 in tables 2 to 5, B2 and B3 (Eq. 11 and Table 6).

Model: RKPR-EOS + CMRs

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